Accelerating Design Exploration and Optimization with Surrogate Physics Models

A Mid Semester Project Report

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by

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Abstract

We present a practical framework for surrogate-modelling-based design optimisation (SMbDO) that addresses the high computational cost of running many high-fidelity simulations during engineering design. As a focused case study, the framework is applied to the inverse design of a gyroid-based heat exchanger — a representative problem that is both relevant to real applications and tractable for demonstration.

The central aim is to maximise the core heat-exchange surface area while respecting strict constraints on mass, pressure drop, and mean flow velocity. To do this we build a physics-informed surrogate consisting of two connected Artificial Neural Networks (ANNs). The architecture deliberately separates geometry-driven predictions (mass and surface area) from flow-driven predictions (pressure drop and average velocity), which improves interpretability and helps the models generalise. Training data are obtained from high-fidelity Computational Fluid Dynamics (CFD) simulations.

Because the surrogate evaluates orders of magnitude faster than CFD, we use a two-stage optimisation: Bayesian Optimisation to explore the global design space efficiently, followed by a gradient-based local refinement for precise tuning. This pipeline finds an optimal combination of lattice cell sizes and inlet velocity that meets all design constraints.

This report is a mid-term proof-of-concept for the SMbDO pipeline. Results indicate the surrogate-driven approach reduces optimisation time from a potentially prohibitive multi-week effort to a matter of hours. Future work will validate the surrogate against additional high-fidelity runs and extend the framework to handle more complex and multi-objective design problems.

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Notation and Abbreviations

C_x Lattice cell size in the X-direction (mm)

C_{yz} Lattice cell size in the Y/Z-directions (mm)

v_{in} Inlet flow velocity (mm/s)

v_{avg} Average flow velocity in the core (mm/s)

Δp Pressure drop across the heat exchanger (Pa)

A_s Core surface area (mm^2)

M Mass of the heat exchanger core (g)

D_h Hydraulic diameter (m)

Re Reynolds number (dimensionless)

Nu Nusselt number (dimensionless)

ANN Artificial Neural Network

CFD Computational Fluid Dynamics

RMSE Root Mean Squared Error

TPMS Triply Periodic Minimal Surface

LPBF Laser Powder Bed Fusion

Chapter 1

Introduction

1.1 Background and motivation

Modern engineering design — across aerospace, automotive, and biomedical domains — is becoming ever more complex and multi-objective. Achieving higher performance under tight constraints on mass, volume, and energy requires accurate prediction of physics such as fluid flow, structural response, and heat transfer. These behaviors are governed by nonlinear partial differential equations and are typically evaluated using high-fidelity numerical methods such as Computational Fluid Dynamics (CFD) and Finite Element Analysis (FEA) (Zienkiewicz et al., 2005; Versteeg and Malalasekera, 2007).

Relying on high-fidelity simulations creates a practical bottleneck: single runs can take hours or days on HPC systems, which conflicts with the iterative nature of design workflows (Bhaskar and Roy, 2017). Full optimisation, sensitivity studies, and design-space exploration demand many function evaluations; when each evaluation is expensive, an optimisation can become infeasible in practice.

Additive Manufacturing (AM) expands design freedom by allowing highly complex internal geometries, such as lattice or TPMS (Triply Periodic Minimal Surface) structures, that were previously impossible or impractical. That increased freedom is a double-edged sword: it enables superior performance but also vastly enlarges the design space, making brute-force or manually guided optimisation impractical (Gibson et al., 2014). The result is a pressing need for methods that retain the benefits of high-fidelity physics models while dramatically reducing the cost per evaluation.

1.2 Surrogate-driven inverse design: a generalizable framework

To resolve this tension between design freedom and computational cost, we adopt a surrogate-model-driven inverse design framework. A surrogate (or metamodel) is a data-driven function that approximates the mapping from design variables to performance metrics. The typical workflow is: run a set of strategically chosen high-fidelity simulations (guided by Design of Experiments), train a machine-learning model (e.g., an ANN) on that data, and then use the trained surrogate to predict performance for new designs at negligible cost.

The main advantage is computational efficiency: the expensive CFD work is performed offline to create the training dataset; once the surrogate is trained, predictions are orders of magnitude faster than running full simulations. This makes inverse design — where one specifies desired performance and uses optimisation to find the design parameters that achieve it — tractable.

This general approach is domain-agnostic. For this project, we apply it to heat exchanger design because heat exchangers are widely used and improvements can yield substantial energy and weight savings. The framework's modular components — parametric geometry, automated simulation, surrogate training, and optimisation — can be transferred to other physics problems.

1.3 Problem structuring and justification

We focus on developing and demonstrating a surrogate-assisted inverse design workflow for a gyroid heat exchanger. The inverse design problem is to find geometric and flow parameters that maximise a performance metric while satisfying engineering constraints.

1.3.1 Gyroid unit cell: shape and mathematical representation

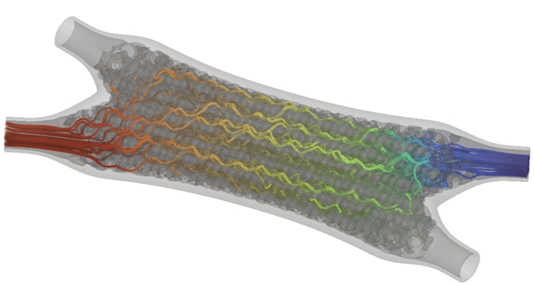
The gyroid belongs to the TPMS family: a smooth, periodic surface without planar symmetry or straight lines. Its topology yields a high surface-to-volume ratio and continuous, tortuous channels that are beneficial for heat transfer and mixing.

The gyroid's implicit equation is:

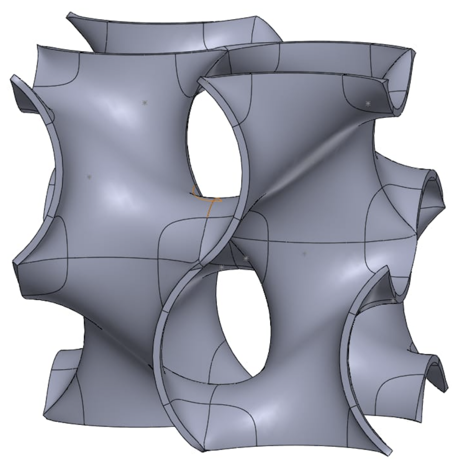
$$\sin\left(\frac{2\pi x}{C_x}\right)\cos\left(\frac{2\pi y}{C_{yz}}\right)+\sin\left(\frac{2\pi y}{C_{yz}}\right)\cos\left(\frac{2\pi z}{C_{yz}}\right)+\sin\left(\frac{2\pi z}{C_{yz}}\right)\cos\left(\frac{2\pi x}{C_x}\right)=0 \qquad (1.1)$$

Here C_x and C_{yz} control the lattice spacing in the x and y/z directions respectively. The zero level set defines the surface; positive and negative values define the two complementary volumes separated by the surface.

Figure 1.1b shows a single gyroid cell. The connected channels avoid dead-ends, reducing flow maldistribution and aiding uniform heat transfer.



(a) The full heat exchanger assembly.



(b) A single unit cell of the gyroid structure.

Figure 1.1: The heat exchanger geometry shown at both the macro and micro scale.

By varying C_x and C_{yz} we directly affect the core's surface area, mass, and the geometry of the flow passages — variables that are central to the optimisation problem.

1.3.2 Objective: maximise surface area

The goal is to maximise the core surface area A_s. For a heat exchanger, the total heat transfer rate is

$$Q = UA_s\Delta T_{LM},$$

where U is the overall heat transfer coefficient and ΔT_{LM} is the log-mean temperature difference (Incropera et al., 2007). For given fluids and operating conditions, U and ΔT_{LM} are approximately fixed, so increasing A_s is the most direct lever to increase heat transfer capacity in a fixed envelope.

1.3.3 Design constraints and their physical significance

To ensure the design is practical and realistic, we impose the following constraints:

- **Mass ($M < 125$ g):** Many applications, especially mobile and airborne systems, are mass-sensitive. A hard mass limit prevents impractically heavy cores.

- **Pressure Drop ($\Delta p < 8000$ Pa):** Pressure drop impacts pumping power ($P_{pump} \propto \Delta p \cdot \dot{V}$) (White, 2011). Excessive pressure loss increases parasitic power and reduces system efficiency. The 8000 Pa cap is representative of allowable limits in compact cooling loops.

- **Average Velocity ($v_{avg} > 520$ mm/s):** A minimum mean flow velocity helps maintain convective heat transfer effectiveness and avoid stagnant regions and thick thermal boundary layers (Incropera et al., 2007).

1.4 Project objectives

The mid-term goals for this work are:

1. Develop a computationally cheap surrogate that predicts the four key outputs (M, A_s, Δp, v_{avg}) from the three inputs (C_x, C_{yz}, v_{in}).

2. Structure the surrogate to reflect relevant physics to improve accuracy and generalisability.

3. Implement an efficient optimisation routine that exploits the surrogate's speed to solve the constrained inverse design problem.

4. Provide a proof-of-concept demonstration; rigorous validation is reserved for the next project phase.

Chapter 2

Literature Review

This chapter reviews background work that underpins our approach. We cover surrogate modelling, optimisation techniques, surrogate-assisted inverse design, and recent work on gyroid-based heat exchangers.

2.1 Surrogate modelling in engineering design

Because high-fidelity simulations are expensive, surrogate models (metamodels) are widely used to approximate costly codes (Forrester et al., 2008). Surrogates learn input-output mappings from a limited set of simulations. Methods range from polynomial response surfaces and Kriging to Radial Basis Functions (RBFs) and Artificial Neural Networks (ANNs) (Bhaskar and Roy, 2017). ANNs are particularly useful for highly nonlinear, high-dimensional problems and have been applied in thermofluid problems to predict quantities such as drag, heat transfer rates, and pressure distributions, often yielding large speed-ups over CFD (Zhang and Liu, 2021).

2.2 Design optimisation techniques

Optimisation algorithms can be gradient-based or gradient-free. Gradient-based methods (e.g., SQP, Adam) are efficient but can become trapped in local optima; gradient-free approaches (e.g., genetic algorithms) explore globally but often need many evaluations (Arora, 2011). Bayesian Optimisation offers a balanced alternative by building a probabilistic surrogate and using an acquisition function to choose informative samples, making it effective for expensive black-box functions (Shahriari et al., 2015).

2.3 Surrogate-assisted inverse design

Inverse design specifies desired performance and searches for the design that achieves it. When each simulation is expensive, inverse design becomes impractical; replacing the simulator with a fast surrogate makes the optimisation loop tractable (Viana et al., 2014). A practical strategy is to use a global search method (e.g., Bayesian Optimisation) on the surrogate, then refine promising solutions with a local, gradient-based method.

2.4 Gyroid heat exchangers: state-of-the-art

TPMS structures like the gyroid are attractive for heat exchangers because of their high surface-area-to-volume ratios and complex internal channels. Kus et al. (Kus et al., 2024) performed numerical and experimental studies on LPBF-manufactured gyroid heat exchangers and validated CFD (k-omega SST) against experiments, finding that cell size strongly controls the trade-off between heat-transfer area and pressure drop. Smaller cells boost surface area but sharply increase pressure loss — a key trade-off our optimisation must manage. Other studies agree that TPMS geometries promote mixing and early transition to turbulence, enhancing convective heat transfer (Kaur and Singh, 2021).

Chapter 3

Formulation and Methodology

This chapter defines the optimisation problem and details the dataset, surrogate architecture, and optimisation algorithms used.

3.1 Problem formulation

We seek the design parameters that maximise the core surface area A_s. The design vector is (C_x, C_{yz}, v_{in}). Formally:

$$\arg\max_{C_x, C_{yz}, v_{in}} \quad A_s(C_x, C_{yz}) \tag{3.1}$$

subject to:

$$M(C_x, C_{yz}) < 125 \text{ g} \tag{3.2}$$

$$\Delta p(C_x, C_{yz}, v_{in}) < 8000 \text{ Pa} \tag{3.3}$$

$$v_{avg}(C_x, C_{yz}, v_{in}) > 520 \text{ mm/s} \tag{3.4}$$

and bounds:

$$10 \text{ mm} \leq C_x \leq 25 \text{ mm} \tag{3.5}$$

$$10 \text{ mm} \leq C_{yz} \leq 25 \text{ mm} \tag{3.6}$$

$$2500 \text{ mm/s} \leq v_{in} \leq 3500 \text{ mm/s} \tag{3.7}$$

The mappings $A_s(\cdot)$, $M(\cdot)$, $\Delta p(\cdot)$, and $v_{avg}(\cdot)$ are unknown and are approximated by the surrogate models.

3.2 Dataset and exploratory analysis

Training data came from a full-factorial Design of Experiments (DoE) where CFD simulations were run across the design grid. An initial exploratory analysis (Figure 3.1) revealed informative correlations.

Key observations:

- **Decoupling of Physics:** Mass (M) and Surface Area (A_s) show no correlation with inlet velocity v_{in}, which is expected since these are geometric properties independent of operating flow.

- **Coupled Flow Dynamics:** Pressure drop (Δp) and average velocity (v_{avg}) correlate strongly with all three inputs, consistent with fluid mechanics where geometry and velocity together determine inertial and viscous effects (e.g., $\Delta p \propto v^2$ in turbulent flow) (White, 2011).

These observations motivated the surrogate architecture described next.

3.3 Physics-guided surrogate model architecture

We use a two-stage, physics-guided surrogate composed of two MLPs to exploit the observed decoupling between geometric and flow-dependent quantities. Separating the tasks reduces complexity and improves model interpretability.

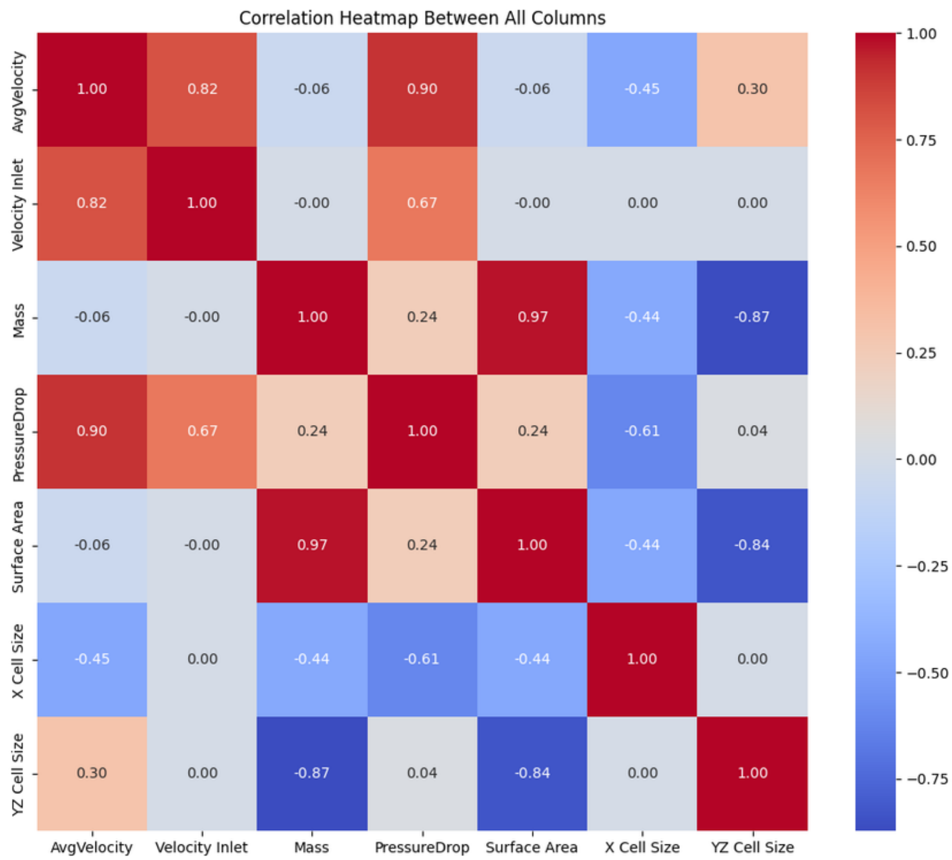


Figure 3.1: Correlation heatmap of the input and output variables. A clear decoupling is visible between the purely geometric properties (Mass, Surface Area) and the inlet velocity (v_{in}), which shows zero correlation.

Figure 3.2 illustrates the architecture: Model 1 predicts geometric outputs, and Model 2 predicts flow outputs using both the original inputs and Model 1's geometric predictions.

- **Model 1 (Geometric Predictor):** An ANN that receives (C_x, C_{yz}) and outputs (M, A_s):

$$(M, A_s) = \texttt{Model 1}(C_x, C_{yz}) \tag{3.8}$$

- **Model 2 (Fluid Dynamics Predictor):** An ANN that predicts $(\Delta p, v_{avg})$ and takes $(C_x, C_{yz}, U_{in}, M, A_s)$ as inputs:

$$(\Delta p, v_{avg}) = \texttt{Model 2}(C_x, C_{yz}, U_{in}, M, A_s) \tag{3.9}$$

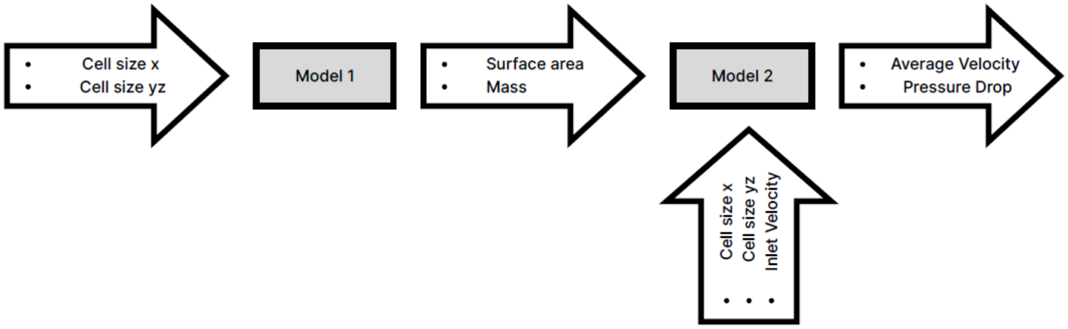


Figure 3.2: The two-stage, physics-informed surrogate model architecture. Model 1 predicts geometric properties, which are then fed into Model 2 along with flow conditions to predict fluid dynamic properties.

This staged design allows each network to focus on a simpler subproblem and embeds physically meaningful quantities (e.g., A_s, M) into the flow predictor.

3.3.1 Physics guidance and rationale

The architecture is grounded in fluid-mechanical reasoning:

1. **Geometric Decoupling:** Cell sizes determine solid volume and wetted surface area. These are geometric invariants that do not depend on inlet velocity, so Model 1 can learn them directly.

2. **Fluid Flow Dependence:** Flow quantities depend on inlet conditions and the geometry-influenced properties. Important derived quantities include:

   - **Hydraulic Diameter (D_h):**

$$D_h = \frac{4V_s}{A_s} \tag{3.10}$$

     where V_s is the fluid volume and A_s the wetted surface area. D_h captures how geometry sets the effective flow passage size.

- **Reynolds Number (Re):**

$$Re = \frac{\rho D_h U_{in}}{\mu} \tag{3.11}$$

  which combines fluid properties, geometry, and velocity to indicate flow regime.

- **Cell Size Aspect Ratio (C_x/C_{yz}):** This ratio controls anisotropy in the structure and affects flow distribution and pressure loss.

Including M and A_s as inputs to Model 2 lets the network implicitly use these physics-derived effects.

Embedding these relationships simplifies learning and yields a more robust surrogate.

3.4 Model implementation and training

Both models were built as Multi-Layer Perceptrons (MLPs) in PyTorch. Data splits were 75% training, 15% validation, and 10% test. Inputs were standardised using 'StandardScaler'. Hyperparameters (Table 3.1) were chosen by a probabilistic search and models were trained for 100 epochs until validation loss converged.

Table 3.1: Hyperparameters of the Neural Network Models.

Parameter	**Model 1 (Geometric)**	**Model 2 (Fluid Dynamics)**
Input Dimension	2	5
Output Dimension	2	2
Hidden Layers	3	3
Neuron Count	[237, 182, 240]	[262, 469, 70]
Activation Function	Leaky ReLU	ReLU
Optimizer	Adam	Adam
Learning Rate	0.0067	0.0024
Batch Size	Full Batch	32
Weight Decay	None	2.939×10^{-5}

3.5 Optimisation strategy

We use a structured two-stage optimisation that leverages the surrogate speed while respecting physics-driven decoupling. Observing that mass is the primary limiting constraint, and both the objective (A_s) and the mass constraint depend solely on geometry, we decouple the problem into a 2D geometric optimisation and a subsequent 1D feasibility search over inlet velocity.

- **Stage 1: Cell Size Optimisation.** Use Model 1 to find (C_x^*, C_{yz}^*) that maximises A_s under the mass constraint using a hybrid optimisation approach.
- **Stage 2: Inlet Velocity Feasibility Search.** With the geometry fixed, use Model 2 to search over v_{in} and identify velocities that satisfy Δp and v_{avg} constraints. The nominal inlet velocity is chosen as the midpoint of the feasible interval.

3.5.1 Global exploration via Bayesian optimisation

We begin with Bayesian Optimisation to identify promising regions in the (C_x, C_{yz}) space.

1. **Surrogate Models (Gaussian Processes):** Two Gaussian Process models are trained: one for surface area A_s and one for mass M, each as a function of (C_x, C_{yz}). We use an RBF kernel. The GP provides mean predictions and predictive uncertainties.
2. **Constrained Acquisition Function:** We employ a Constrained Expected Improvement (cEI) acquisition:

$$\mathrm{cEI}(x) = \mathrm{EI}(x) \times P(M(x) \leq M_{\mathrm{limit}}) \tag{3.12}$$

   - **Expected Improvement (EI):** Favors locations likely to improve the best feasible solution, balancing exploitation and exploration.

- **Probability of Feasibility (P):** Computed from the GP for mass; it down-weights regions likely to violate the mass constraint.

Iteratively, the algorithm selects the point maximizing cEI, evaluates it using Model 1, and updates the GPs.

3.5.2 Local refinement via gradient ascent

After Bayesian Optimisation narrows the search, we refine the best candidate with gradient ascent using the differentiable PyTorch Model 1.

1. **Direct Gradient Calculation:** Compute ∇A_s with backpropagation to follow the steepest ascent direction.
2. **Constrained Optimisation with a Penalty Function:** We minimise a loss constructed as:
$$\text{Loss} = -A_s + w \times \text{ReLU}(M - M_{\text{limit}}) \tag{3.13}$$
Minimising this loss maximises A_s while penalising mass violations; w is a penalty weight tuned as a hyperparameter. Adam is used as the optimizer.
3. **Bound Enforcement:** Parameter values are clamped to the physical bounds [10, 25] mm at each step to keep the solution feasible.

This refinement yields a precise local optimum within the promising region discovered by the global search.

3.5.3 Inlet velocity feasibility search

Fix (C_x^*, C_{yz}^*) and sweep v_{in} over [2500, 3500] mm/s. For each v_{in}, Model 2 predicts v_{avg} and Δp. The subset of v_{in} values that meet both constraints defines the feasible operating range; we select the midpoint of that range as the nominal inlet velocity.

Chapter 4

Results and Discussion

This chapter summarises surrogate performance and reports optimisation outcomes.

4.1 Surrogate model performance

Models were evaluated on the held-out test set. Results in Table 4.1 show high predictive fidelity.

Table 4.1: Performance of Surrogate Models on the Test Set.

Model / Output	**RMSE**	R^2 **Score**
Model 1 (Geometric)		
Mass (g)	0.146	1.000
Surface Area (mm^2)	206.534	0.996
Model 2 (Fluid Dynamics)		
Avg. Velocity (mm/s)	9.255	0.982
Pressure Drop (Pa)	489.360	0.960

All R^2 values exceed 0.96 and RMSEs are small relative to the output scales, indicating the surrogates capture the dominant behavior well and are suitable for optimisation tasks.

4.2 Optimisation results

The two-stage optimisation found the final design reported in Table 4.2.

Table 4.2: Optimal Design Parameters and Predicted Performance.

Parameter	**Optimal Value**	**Constraint / Target**
Inputs (Design Variables)		
Cell Size X (C_x)	21.787 mm	[10, 25]
Cell Size YZ (C_{yz})	20.529 mm	[10, 25]
Inlet Velocity (v_{in})	3168.3 mm/s	[2500, 3500]
Outputs (Performance Metrics)		
Surface Area (A_s)	**23559.94 mm^2**	Maximised
Mass (M)	124.99 g	$<$ 125 g
Pressure Drop (Δp)	6772.29 Pa	$<$ 8000 Pa
Avg. Velocity (v_{avg})	566.02 mm/s	$>$ 520 mm/s

The solution sits right at the mass limit ($M \approx 125$ g), which aligns with the intuition that increasing surface area tends to increase mass; the optimiser leverages available mass up to the constraint to maximise A_s. Pressure drop and average velocity constraints are satisfied.

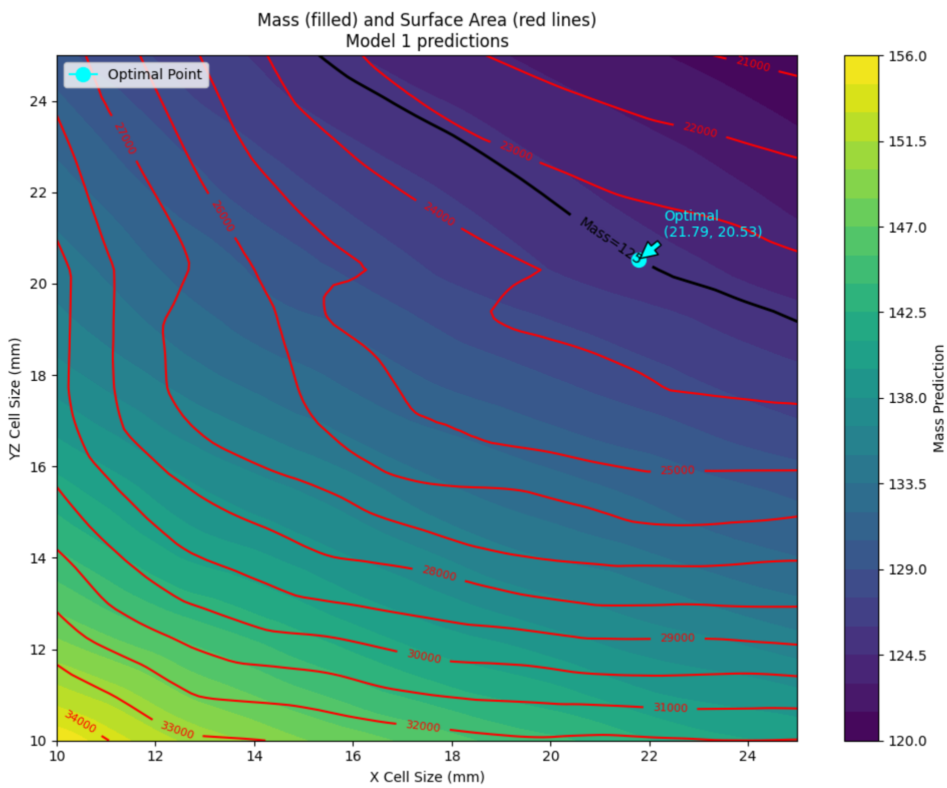


Figure 4.1: Heatmap of predicted mass (color-filled) and surface area (red contour lines) for varying X and YZ cell sizes, as estimated by Model 1.

Chapter 5

Summary and Conclusions

We developed a surrogate-assisted inverse design framework for a gyroid-based heat exchanger. The main objective was to maximise heat-exchange surface area under practical constraints on mass, pressure drop, and mean flow velocity.

A physics-informed surrogate split into two coupled ANNs was trained to predict geometric and flow-dependent metrics. The surrogate achieved R^2 scores above 0.96 and gives near-instantaneous predictions compared to CFD. A two-stage optimisation — Bayesian global search followed by gradient-based local refinement — found a design achieving approximately 23,560 mm^2 of surface area while satisfying all constraints.

This mid-term work shows that combining physics-guided machine learning with modern optimisation dramatically accelerates design exploration for complex thermofluid systems. The approach offers a practical tool for engineers to explore large design spaces quickly; further validation and extensions to multi-objective and robust optimisation are planned.

Chapter 6

Future Work

Key directions for future development include:

- **Advanced Data Generation and Validation Strategies:** The immediate next step is to run a high-fidelity simulation at the surrogate-predicted optimal point to validate surrogate accuracy. We will also explore active learning, where the surrogate suggests new simulations in regions of high uncertainty or promising performance, and alternative DoE strategies (e.g., Sobol sequences, optimized Latin Hypercube) to improve data efficiency.

- **Vibroacoustics and Noise Control:** The framework can be extended to inverse-design problems in vibroacoustics, where balancing structural robustness with acoustic performance is critical. Surrogate models would enable topology and parameter searches that reduce radiated noise without adding heavy damping treatments.

- **Uncertainty Quantification for Robust Design:** Real designs face manufacturing tolerances, material variability, and varying operating conditions. Using the fast surrogate, we can propagate input uncertainties to quantify performance distributions and then optimise for robustness.

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